

# Joshua Cook - ENGINEERING PHYSICS

[Joshcook452@yahoo.com](mailto:Joshcook452@yahoo.com) | [Portfolio](#) | 604-815-7883

This resume highlights my work history and my discrete hardware project. To see my other recent PCB, firmware and CAD projects please check out [the projects on my portfolio](#).

## SKILLS

---

**Embedded Hardware & PCB Design:** KiCad (4-Layer Mixed-Signal, EMI Mitigation), Schematic Capture, STM32F4, Audio ADCs, Bluetooth RF Transceivers.

**Firmware & Low-Level Protocols:** C/C++, I2S, 4-Wire UART (HCI), SPI, DMA Data Pipelines, Closed-Loop PID Control.

**Prototyping, CAD & Rework:** Fusion 360 (Parametric & Sheet Metal), SMT Reflow Soldering, Fine-Pitch SMD Rework (Hot Air/Iron), 3D Printing.

## EXPERIENCE AND OTHER TECHNICAL PROJECTS

---

### 8-bit breadboard computer

**Oct. 2024 – Jun. 2025**

- Architected an 8-bit breadboard computer using discrete 7400-series logic gates, EEPROMs, registers, and SRAM, incorporating memory-mapped I/O with 8 GPIO pins.
- Simulated the full CPU architecture in a logic gate simulator prior to hardware assembly to validate clock timing, bus control signals, and instruction cycles.
- Designed a flexible EEPROM-based microcode instruction decoder, enabling custom instruction set architecture (ISA) modifications directly through software firmware updates.
- Assembled and debugged the physical hardware on solderless breadboards using a Digital Multimeter (DMM) to verify power rails and shorts, alongside a low-frequency clock and LED indicators for step-by-step logic tracing.

### Painter

**Jun. 2026 – Aug. 2026**

Magellan Coatings, Squamish

- Collaborated within multi-disciplinary teams of 2–5 workers across 14+ job sites to complete surface restoration, renovation, and new construction finishing projects.
- Prepared diverse substrates, including drywall, raw timber, finished woodwork, baseboards, and meta, by pressure washing, mudding, sanding, chemical stripping, and taping.
- Erected temporary job-site structures, including multi-tier scaffolding systems and protective containment barriers, enforcing site safety protocols during high-elevation operations.

- Executed precise surface coating applications, managing material curing conditions and application methods (cutting, rolling, and staining) to ensure structural adhesion and durability.

**PhotoLab and Tech Specialist****Aug. 2024 – Sep. 2025**

London Drugs, Squamish

- Delivered exceptional customer service in PhotoLab and Tech Specialist roles, earning store manager recognition for personalized assistance and technical product guidance.
- Diagnosed and maintained high-volume printing and scanning equipment during live operations, minimizing machinery downtime through proactive troubleshooting and repair.
- Streamlined order workflows by sorting and prioritizing incoming requests based on turnaround times and job volume to guarantee efficient, on-time delivery.
- Optimized workspace organization and hardware layout in collaboration with the lab manager, receiving praise from store management for significantly improving operational cleanliness.

**EDUCATION**

---

**Engineering Physics, BSc. (in-progress)**

UBC, Vancouver